

❖ Problems: Formulation of LPP

1. A company makes two products tables, chairs which must be preceded to assembly and finishing department. Assembling has 60hrs available per week, finishing can handle up to 48hrs per week. To manufacture a table it requires 4hours of Assembly, 2hours of Finishing. To manufacture a chair it requires 2hours of Assembly, 4hours of Finishing. If the profit is Rs.8 per table and Rs.6 per chair, the problem is to determine the best possible combination of tables and chairs to maximize the profit.
2. A company produces 3 products p, q and r from 3 raw materials A, B and C. One unit of 'p' requires 2 units of 'A' and 3 units of 'B'. One unit of 'q' requires 2 units of 'B' and 5 units of 'C'. One unit of 'r' requires 3 units of 'A', 2 units of 'B' and 4 units of 'C'. A company having 8 units of 'A', 10 units of 'B' and 15 units of 'C'. Profits per unit of product p, q, r are Rs.3, Rs.4, Rs.5 respectively. Formulate the problem mathematically to maximize the product profit.
3. Vitamins B1 and B2 are found in two foods F1 and F2. 1 unit of F1 contains 3 units of B1 and 4 units of B2. 1 unit of F2 contains 5 units of B1 and 3 units of B2 respectively. Minimum daily prescribed consumption of B1 & B2 is 50 and 60 units respectively. Cost per unit of F1 & F2 is Rs. 6 & Rs. 3 respectively. Formulate as LPP.
4. A company is making 2 products A, B. The cost of producing one unit of A, B is Rs.60, Rs.80 respectively. As per the agreement the company must supply at least 200 units of product 'B' to the regular customers. One unit of 'A' requires 1 machine hour. The total number of machine hours required by 'A' are 400 machine hours. 'B' does not require any machine hours. One unit of A, B requires labour hour of 1 hour each and a total of 500 labour hours are available. The company wants to minimize the cost of production by satisfying given requirement. Formulate the problem as linear programming problem.
5. An animal feed company must produce 200 kg of a mixture consisting of ingredients A and B. The ingredient A costs Rs. 3 per Kg. and B cost Rs. 5 per Kg. Not more than 80 Kg. of A can be used and at least 60 Kg. of B must be used. Formulate the problem as LPP.
6. A firm makes two products A and B, and has a total production capacity of 9 tonnes per day, A and B requires the same production capacity. The firm has a permanent contract to supply at least 2 tonnes of A and at least 3 tonnes of B per day to another company. Each tonne of A requires 20 machine hours production time and each tonne of B requires 50 machine hours production time. The daily maximum possible number of machine hours is 360. All the firm's output can be sold, and the profit made is Rs. 60 per tonne of A and 120 per tonne of B. Formulate the problem as LPP to maximize the profit.
7. A firm produces three products. These products are processed on three different machines. The time required to manufacture one unit of each of the three products and the daily capacity of the three machines are given in the table below: It is required to determine the daily number of units to be manufactured for each product. The profit per unit for product 1, 2 and 3 is Rs. 4, Rs.3 and Rs.6 respectively. It is assumed that all the amounts produced are consumed in the market. Formulate the mathematical (L.P.) model that will maximise the daily profit.

Machine	Time per unit (Minutes)			Machine Capacity (minutes/day)
	Product 1	Product 2	Product 3	
M ₁	2	3	2	440
M ₂	4	-	3	470
M ₃	2	5	-	430

8.

A firm is engaged in animal breeding. The animals are to be given nutrition supplements everyday. There are two products A and B which contain the three required nutrients.

Nutrients	Quantity/unit		Minimum Requirement
	A	B	
1	72	12	216
2	6	24	72
3	40	20	200

Product cost per unit are : A, Rs. 40. and B, Rs. 80. Formulate the problem as LPP.

❖ Graphical Method: Problems

Solve the following LP Problems by graphical method.

1. Maximize $Z = 8x_1 + 6x_2$

Subject to the constraints:

$$4x_1 + 2x_2 \leq 60$$

$$2x_1 + 4x_2 \leq 48 \text{ and}$$

$$x_1, x_2 \geq 0$$

2. Minimize $Z = 60x_1 + 80x_2$

Subject to the constraints:

$$x_2 \geq 200$$

$$x_1 \leq 400$$

$$x_1 + x_2 \leq 500 \text{ and}$$

$$x_1, x_2 \geq 0$$

3. Minimize $Z = 20x_1 + 40x_2$

Subject to the constraints:

$$36x_1 + 6x_2 \geq 108$$

$$3x_1 + 12x_2 \geq 36$$

$$20x_1 + 10x_2 \geq 100 \text{ and}$$

$$x_1, x_2 \geq 0$$

❖ Special cases of Graphical Method:

1. Infeasibility

Maximize $Z = x_1 + x_2$

Subject to the constraints:

$$2x_1 + 2x_2 \leq 1$$

$$3x_1 + x_2 \geq 3 \text{ and}$$

$$x_1, x_2 \geq 0$$

2. Unboundedness

Maximize $Z = 10x_1 + 5x_2$

Subject to the constraints:

$$x_1 \geq 5$$

$$x_2 \leq 10$$

$$x_1 + 2x_2 \geq 10 \text{ and}$$

$$x_1, x_2 \geq 0$$

3. Redundancy

Maximize $Z = 3x_1 + 6x_2$

Subject to the constraints:

$$x_1 + x_2 \leq 20$$

$$x_1 \leq 35$$

$$2x_1 + x_2 \leq 30 \text{ and}$$

$$x_1, x_2 \geq 0$$

4. Alternative optimum solution

Maximize $Z = 8x_1 + 4x_2$

Subject to the constraints:

$$4x_1 + 2x_2 \leq 60$$

$$2x_1 + 4x_2 \leq 48 \text{ and}$$

$$x_1, x_2 \geq 0$$